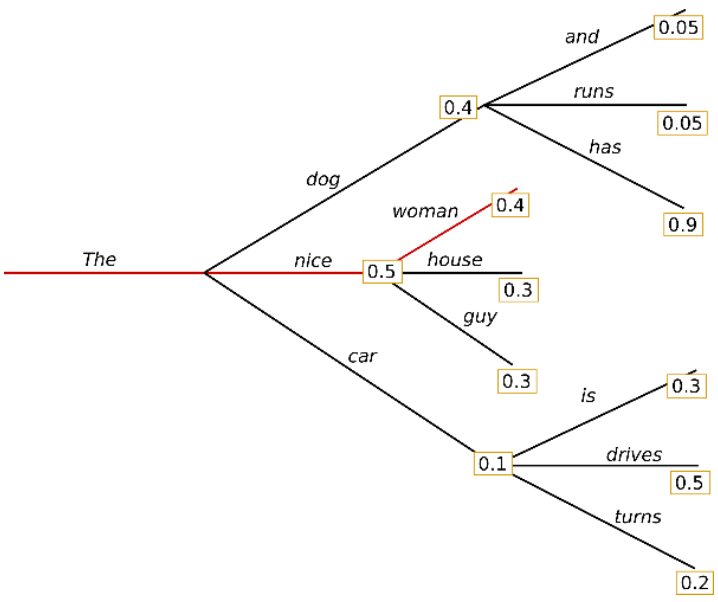
Deep Learning for NLP

Decoding Strategies
Stochastic Decoding & CS/CD



Learning goals

- Get to know different stochastic decoding strategies
- Learn about sampling with temperature, top-k sampling and top-p (nucleus) sampling
- learn about contrastive search and contrastive decoding

SAMPLING MOTIVATION I

- *Creativity and Variation*: Sampling methods produce varied outputs for the same input, useful in creative applications like story generation and dialogue systems.
- *Avoiding Repetition*: These methods are less likely to generate repetitive loops compared to deterministic methods.

SAMPLING (WITH TEMPERATURE) (1) I

The next token is selected randomly based on its conditional probability distribution. To control the randomness of the output sequence, a temperature parameter can be applied to the softmax function

$$\sigma\left(z_i\right) = \frac{e^{\frac{z_i}{temp}}}{\sum_{j=1}^{N} e^{\frac{z_j}{temp}}}$$

- $temp \to \infty$: Output distribution $\approx$ Uniform distribution
- $temp \to 0$: Output distribution $\approx$ Point mass (Greedy search)

SAMPLING (WITH TEMPERATURE) (2) I

Prompt: "Once upon a time"

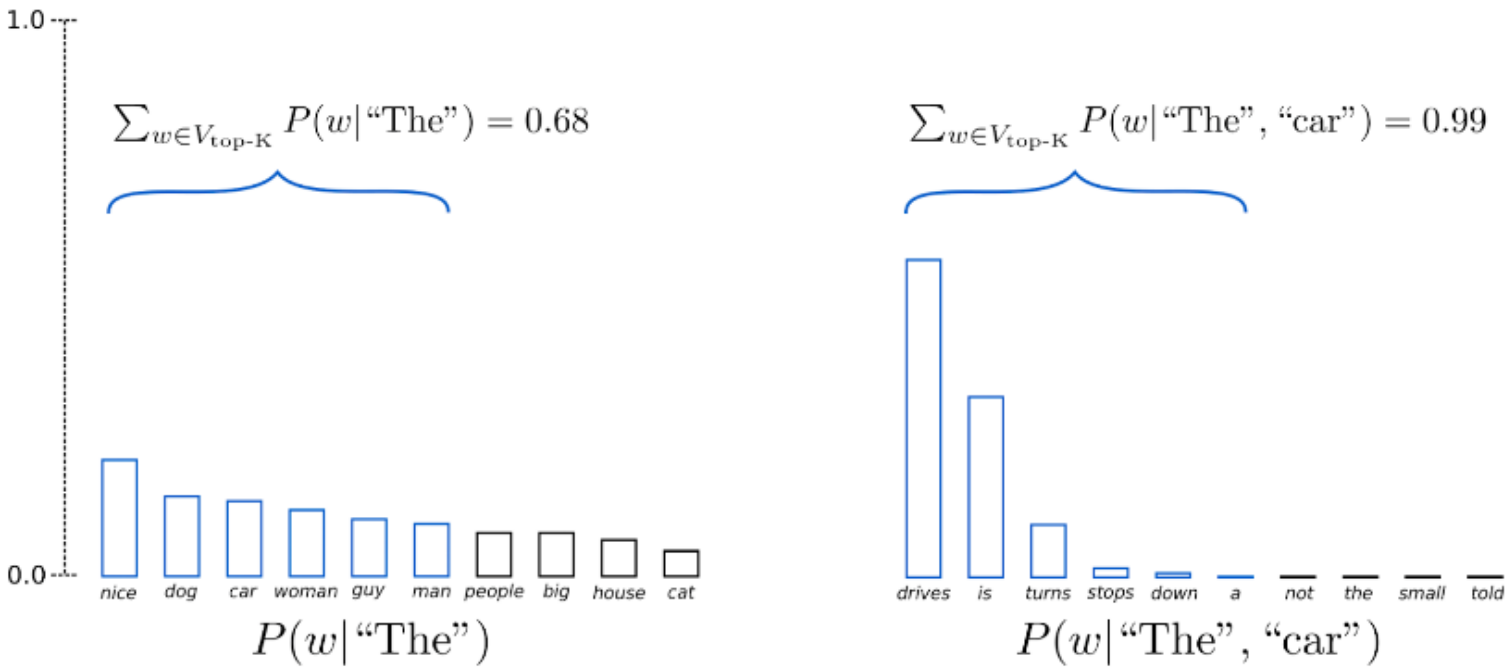
- Sampling with low temperature: *", during the Second World War, during the final months for his three most talented young players, the coach, Harry Gregg said this"*
- Sampling with high temperature: *"— well. Nowhere you call back my call, not on time; never the two on account my four. Do not come." This old woman — you might have liked, she herself — she did smile."*

The generated stories are diverse but sometimes very erratic.

$\Rightarrow$ Sample from the top-k tokens

TOP-K SAMPLING ▸ Fan et al., 2018

In Top-k sampling, the k most likely next tokens are filtered, and the probability mass is redistributed. Visualization for $k = 6$ in two sampling steps:



▸ huggingface, Patrick von Platen

TOP-K SAMPLING I

Prompt: "Once upon a time"

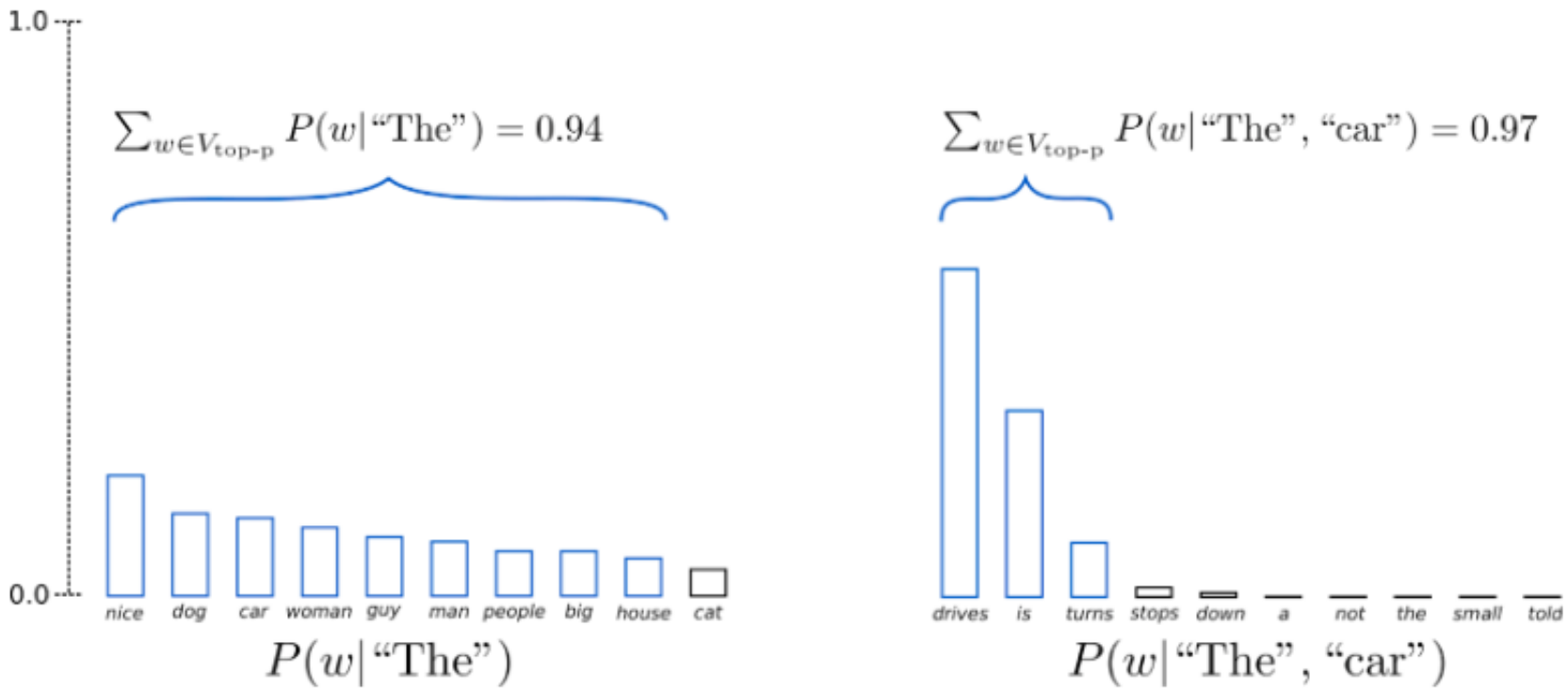
- Top-k , $k = 100$: *"when I was young the internet was a mysterious landscape full of new and exciting ideas. I read ebooks, watched videos, read short stories"*

The quality has improved, but the fixed k might be counterproductive

$\Rightarrow$ Make k dynamic

TOP-*P* (NUCLEUS) SAMPLING ▸ Holtzman et al., 2019

Top-*p* sampling chooses from the smallest possible set of tokens whose cumulative probability exceeds the probability threshold *p*. The probability mass is then redistributed accordingly. Visualization with a threshold $p = 0.92$:



▸ huggingface, Patrick von Platen

TOP-P (NUCLEUS) SAMPLING I

Prompt: "Once upon a time"

- Top-p , $p = 0.92$: *"there were four major political parties in the United States. Since then, however, they have become even more of a novelty. For the past few decades, there have been only two."*

SOTA for many years, default decoding strategy in various GPT versions, but sometimes erratic depending on p and the sampled tokens.

Question: Can there be a balance of coherence and diversity?
$\Rightarrow$ Contrastive search

CONTRASTIVE SEARCH ▸ Su et al., 2022

$$x_t = \arg\max_{v \in V^{(k)}} \left\{ (1-\alpha) \times \underbrace{p_\theta(v|\boldsymbol{x}_{<t})}_{\text{model confidence}} - \alpha \times \underbrace{(\max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\})}_{\text{degeneration penalty}} \right\}$$

When generating output, contrastive search jointly considers:

- The probability predicted by the language model to maintain the semantic coherence between the generated text and the prompt.
- The similarity with respect to the previous context to avoid degeneration (as in Greedy or Beam search)

$\Rightarrow$ An "ideal" token should have a high probability and bring diversity to the story.

Empirical studies suggest $k \in \{5, 8, 10, 15\}$ and $\alpha \in \{0.4, 0.5, 0.6\}$

▸ Su & Collier, 2023 ▸ Su & Xu, 2022 ▸ Su et al., 2022

CONTRASTIVE SEARCH FORMULA I

▸ huggingface, Tian Lan

Let's have a closer look at the formula for Contrastive Search:

$$x_t = \underset{v \in V^{(k)}}{\textit{argmax}} \left\{ (1-\alpha) \times p_\theta(v|\mathbf{x}_{<t}) - \alpha \times (max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\}) \right\}$$

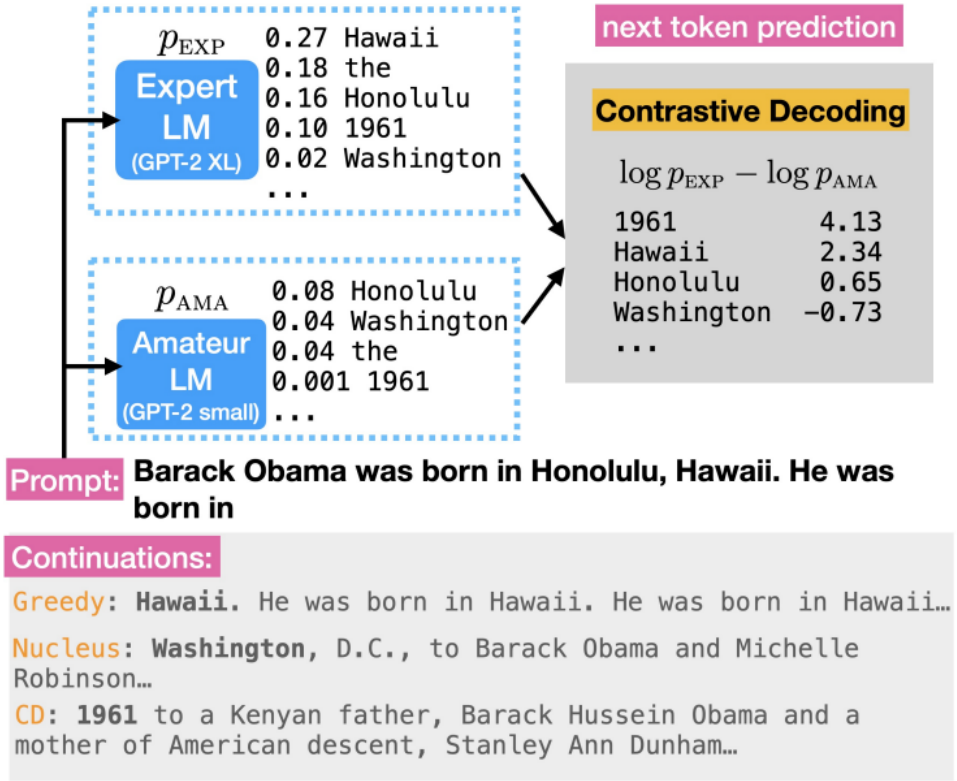
- x_t is the output token and $\mathbf{x}_{<t}$ the context
- $V^{(k)}$ is the set of top-k predictions from the models probability distribution (this is the same k as in the top-k sampling from earlier)
- $p_\theta(v|\mathbf{x}_{<t})$, the *model confidence*, is the probability of a candidate token v given the context
- $max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\}$, the *degeneration penalty*, measures how similar v is to the context, $s()$ is the cosine similarity between the token representations

CONTRASTIVE SEARCH FORMULA II

- The degeneration penalty is defined as the maximum cosine similarity between the token representation of v, i.e h_v, and of all tokens in the context $\mathbf{x}_{<t}$
- h_v is computed by the language model given the concatination of v and $\mathbf{x}_{<t}$
- In order to maximize the formula we want v to have a high probability and a low degeneration penalty
- Intuitively, a larger degeneration penalty of v means it is more similar (in the representation space) to the context, therefore more likely leading to the problem of model degeneration
- α determines how much weight to give to each component
- For $\alpha = 0$ we only consider the probability and contrastive search becomes greedy search

CONTRASTIVE DECODING

▸ Li et al., 2023



- Contrastive decoding exploits the contrasts between expert and amateur LM of different sizes by choosing tokens that maximize their log-likelihood difference (read the paper if you are interested, not going into more detail here!)